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Week 6: Image Processing

📄 Reading: Week 6 Lecture (30 min)

📄 Ungraded Practice: 6.1 Overview of Image Processing (8 min)

📄 Ungraded Practice: 6.2 Fourier Transforms (17 min)

📄 Practice Assignment: 6.2 Fourier Transforms Self-check Quiz (Submitted)

📄 Reading: 6.3 Video: Convolutions (10 min)

📄 Ungraded Practice: 6.3 Convolutions: Theorems (8 min)

📄 Practice Assignment: 6.3 Convolutions: Theorems Self-check Quiz (Submitted)

📄 Ungraded Practice: 6.4 Image Filtering in Frequency Domain (14 min)

📄 Practice Assignment: 6.4 Image Filtering in Frequency Domain Self-check Quiz (Submitted)

📄 Ungraded Practice: 6.5 Deconvolution (17 min)

📄 Practice Assignment: 6.5 Deconvolution Self-check Quiz (Submitted)

📄 Reading: 6.6 Video: Convolutions (10 min)

📄 Ungraded Practice: 6.6 Sampling Theory and Aliasing (13 min)

📄 Practice Assignment: 6.6 Sampling Theory and Aliasing Self-check Quiz (Submitted)

🗨️ Discussion Prompt: Week 6 Questions and Feedback

Week 6 Quiz

Post Course Survey

Your grade: 100%

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6.6 Sampling Theory and Aliasing Self-check Quiz

1 / 1 point

1. How many chip designs do you need to design?

☐ Squared eye layer/print sensor acting as low-pass filter

☐ Using optical low-pass filter of chip

☒ Both options 1 and 2

☐ None of the above

Ready to review what you've learned before starting the assignment? I'm here to help.

🔗 Help me practice

🗨️ Let's chat

📄 Correct

Please refer to lecture videos.

Assignment details

Due

Nov 1, 11:59 PM CST

Attempts

Unlimited

Try again

2. Which of the following is a valid sampling frequency for a square wave of 100 Hz?

1 / 1 point

☐ 50 Hz

☐ 100 Hz

☐ 200 Hz

☒ None of the above

Your grade

To pass you need at least 50%. We keep your highest score.

View submission

See feedback

📄 Correct

To perfectly be able to reconstruct a signal sampling frequency should be twice that of the highest frequency component. Square waves have frequency components all the way up to infinity, but finite sampling frequency can perfectly reconstruct a square wave.

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